

Midterm 1 Practice Set

A 90-minute closed-book mock exam, complete solutions, correction audit, and three-day study plan

EXAM

Thursday, 1 October 2026, in class

PREPARED

28 September 2026

FORMAT

100 points; 90 minutes; pencil-and-paper only

SOURCE-CONTROLLED EXAM BRIEF

Carlos Fernandez-Granda's 23 September announcement states that Midterm 1 is on Thursday, 1 October, in class; students should arrive at least ten minutes early. The exam covers everything taught through Bayesian statistics, including class examples, Homework 1–3, and the posted practice problems. There are no coding questions, and no materials or devices are permitted.

HOMEWORK 3 PROBLEM 2 – CONTROLLING CORRECTION

The 25 September correction replaces the Second District poll counts with **Trump 210 and Harris 270**. Any calculation using 347 and 286 is superseded. This packet uses 210 and 270 throughout.

How to use this packet

1. Print Part I and solve it in one uninterrupted 90-minute sitting. Do not consult Part II.
2. Grade by derivation, not merely by final answer. Record the first invalid step in every missed part.
3. Use the three-day plan to repair only the chains that broke, then re-solve those parts from a blank page.

Coverage Map and Attack Order

The announcement gives a broad boundary rather than a narrow list. The mock therefore samples every high-yield derivation thread represented by the earlier practice work and Homework 1–3, while keeping every task solvable by hand.

Thread	Required operation	Location
Empirical models, tails, KL	Write the support; distinguish an observed zero from an impossible event; turn average log likelihood into a KL objective.	Problem 1
Likelihood and Pareto MLE	Take logs, differentiate on the valid parameter domain, and verify curvature or a boundary.	Problem 1
Sufficiency, Rao–Blackwell	Factor the likelihood; condition a rough unbiased estimator on the sufficient statistic.	Problem 2
Fisher information, CRLB	Compute the score and information; compare the estimator variance with the lower bound.	Problem 2
Conditional Gaussian	Partition the covariance matrix; apply the Schur-complement formula.	Problem 3
Beta–Binomial updating	Treat prior parameters as pseudo-counts; update successes and failures separately.	Problem 4
Gamma–Poisson prediction	Use the rate parameterization; integrate the Poisson likelihood against the Gamma posterior.	Problem 5

THE FIVE-LINE DERIVATION DISCIPLINE

For any inference problem: **(1)** state the model and support; **(2)** isolate the parameter-dependent kernel; **(3)** name the statistic or conjugate family; **(4)** compute the requested estimator, bound, or posterior; **(5)** interpret the result in the model’s own terms. This order prevents most lost points.

Timing checkpoint

At minute 35 you should have finished Problems 1–2. At minute 50 you should have finished Problem 3. Reserve minutes 50–74 for the corrected poll problem and minutes 74–90 for posterior prediction and review.

Part I: Timed Mock Exam

EXAM CONDITIONS

Time: 90 minutes. Total: 100 points. Use no notes, textbook, calculator, phone, computer, or AI system. Show the mathematical chain that supports each answer. A symbolic beta-tail expression is acceptable; the one normal-CDF value needed for a numerical approximation is supplied in Problem 4.

PROBLEM 1 Pareto likelihood, sufficiency, and KL

20 pts · 17 min

Let $X_1, \dots, X_n$ be i.i.d. Pareto observations with known threshold $m > 0$ and shape $\alpha > 0$:

$$f(x \mid \alpha) = \alpha m^\alpha x^{-(\alpha+1)} \mathbf{1}\{x \geq m\}.$$

- (a) Write the likelihood and log likelihood, derive the maximum-likelihood estimator of α , and verify that your stationary point is the unique maximum. [7 pts]
- (b) Use the factorization theorem to identify a one-dimensional sufficient statistic for α . [4 pts]
- (c) For a separate finite-outcome model p_θ and empirical pmf $\hat{p}$, prove that maximizing the sample log likelihood is equivalent to minimizing $D(\hat{p} \| p_\theta)$. State the exact identity. [6 pts]
- (d) Give one precise reason that an empirical pmf based on a small sample is unreliable for extreme-tail prediction. [3 pts]

PROBLEM 2 Poisson sufficiency, information, and Rao–Blackwell

20 pts · 18 min

Let $X_1, \dots, X_n$ be i.i.d. $\text{Poisson}(\lambda)$, $\lambda > 0$, and let $T = \sum_{i=1}^n X_i$.

- (a) Show that T is sufficient for λ . [4 pts]
- (b) Find the MLE $\hat{\lambda}$ and show that it is unbiased. [4 pts]
- (c) Compute the Fisher information $I_n(\lambda)$, state the Cramér–Rao lower bound for unbiased estimators of λ , and determine whether $\hat{\lambda}$ attains it. [7 pts]
- (d) Starting with the unbiased estimator X_1 , compute $\mathbb{E}[X_1 \mid T]$. Explain what Rao–Blackwellization has done. [5 pts]

PROBLEM 3 Conditional Gaussian calculation

16 pts · 15 min

Suppose (Y, X_1, X_2) is jointly Gaussian with

$$\mathbb{E} \begin{pmatrix} Y \\ X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 10 \\ 2 \\ -1 \end{pmatrix}, \quad \text{Cov} \begin{pmatrix} Y \\ X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 9 & 2 & 1 \\ 2 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

- Find the conditional distribution of Y given $X_1 = 4$ and $X_2 = 0$. Show both the conditional mean and conditional variance. [8 pts]
- What fraction of the marginal variance of Y is explained by conditioning on (X_1, X_2) in this model? [4 pts]
- In one paragraph, explain why a five-dimensional central slice containing no observations cannot support a local nonparametric conditional estimate, while a fitted Gaussian conditional distribution can still be evaluated there. [4 pts]

PROBLEM 4 Corrected Second District Bayesian update

24 pts · 24 min

USE THE CORRECTED POLL COUNTS

2020 two-party totals in Nebraska's Second District: Trump 154,377; Biden 176,468. Build a Beta prior centered at the 2020 Trump share with effective sample size 50. The corrected 2024 poll is **Trump 210, Harris 270**.

Let p be Trump's latent two-party share.

- Find the prior parameters α_0 and β_0 , first as formulas and then to four decimals. [5 pts]
- Derive the posterior distribution of p under binomial sampling. [5 pts]
- Find the posterior mean, the posterior effective sample size, and the posterior-predictive probability that one additional sampled voter supports Trump. [5 pts]
- Write $P(p > 0.5 \mid \text{data})$ exactly as a Beta-tail expression. Then use posterior sd = 0.02154 and $\Phi(2.77) = 0.9972$ to obtain a normal approximation. [5 pts]
- The superseded counts 347 and 286 would give posterior mean 0.5422. Explain, without recomputing everything, why the correction reverses the district-level conclusion. [4 pts]

PROBLEM 5 Gamma–Poisson posterior prediction

20 pts · 16 min

Use the shape–rate convention. Let $\lambda \sim \text{Gamma}(a = 2, b = 1)$, and conditional on λ let $X_1, \dots, X_4$ be i.i.d. $\text{Poisson}(\lambda)$. The observations are 4, 2, 3, 5, so $\sum_i X_i = 14$.

- Derive the posterior distribution of λ . [5 pts]
- Find the posterior mean and posterior mode. [4 pts]
- Integrate out λ to derive the posterior-predictive pmf of a new count Y . Identify the resulting family and its parameters. [7 pts]

- (d) Find $\mathbb{E}[Y \mid \text{data}]$ and $\text{Var}(Y \mid \text{data})$. Explain in one sentence why the predictive variance exceeds the predictive mean. **[4 pts]**

STOP. Record your finish time and confidence (0–3) beside each problem. Do not open Part II until the 90 minutes have elapsed.

Part II: Complete Solutions and Grading Guide

Award points for valid reasoning. A correct final number reached from an invalid probability model should not receive full credit; a correct symbolic setup with a small arithmetic slip should retain most method points.

SOLUTION 1 Pareto likelihood, sufficiency, and KL

(a) For observations $x_i \geq m$,

$$L(\alpha) = \alpha^n m^{n\alpha} \prod_{i=1}^n x_i^{-(\alpha+1)}, \quad \ell(\alpha) = n \log \alpha + n\alpha \log m - (\alpha + 1) \sum_{i=1}^n \log x_i.$$

The score is

$$\ell'(\alpha) = \frac{n}{\alpha} + n \log m - \sum_{i=1}^n \log x_i.$$

Setting it to zero gives

$$\hat{\alpha} = \frac{n}{\sum_{i=1}^n \log(x_i/m)}.$$

Since $\ell''(\alpha) = -n/\alpha^2 < 0$ for every $\alpha > 0$, the log likelihood is strictly concave and the stationary point is the unique global maximum. **[7 pts]**

(b) The likelihood factors into an α -dependent function of

$$T = \sum_{i=1}^n \log(X_i/m)$$

times a factor independent of α . Thus T is sufficient by the factorization theorem. **[4 pts]**

(c) Let $\hat{p}(a)$ be the empirical relative frequency on the finite outcome set $\mathcal{A}$. Then

$$\frac{1}{n} \ell(\theta) = \sum_{a \in \mathcal{A}} \hat{p}(a) \log p_\theta(a) = -H(\hat{p}) - D(\hat{p} \| p_\theta).$$

The empirical entropy $H(\hat{p})$ does not depend on θ , so

$$\arg \max_{\theta} \ell(\theta) = \arg \min_{\theta} D(\hat{p} \| p_\theta).$$

[6 pts]

(d) An empirical pmf assigns probability zero to every unobserved value, including possible values beyond the sample maximum. With a small sample, that zero records absence from the sample rather than impossibility, so the empirical model cannot extrapolate an extreme tail. **[3 pts]**

Common loss

Dropping the support indicator before confirming $x_i \geq m$, or writing the KL identity with the sign reversed.

SOLUTION 2 Poisson sufficiency, information, and Rao–Blackwell

(a) The joint pmf is

$$f(\mathbf{x} \mid \lambda) = \frac{e^{-n\lambda} \lambda^T}{\prod_{i=1}^n x_i!}.$$

All λ -dependence enters through T , so T is sufficient.

[4 pts]

(b)

$$\ell(\lambda) = -n\lambda + T \log \lambda + \text{constant}, \quad \ell'(\lambda) = -n + \frac{T}{\lambda}.$$

Therefore $\hat{\lambda} = T/n = \bar{X}$. Since $\mathbb{E}[T] = n\lambda$, $\mathbb{E}[\hat{\lambda}] = \lambda$.

[4 pts]

(c) For one observation, the score is $X/\lambda - 1$, and

$$I_1(\lambda) = \text{Var}\left(\frac{X}{\lambda} - 1\right) = \frac{1}{\lambda}, \quad I_n(\lambda) = \frac{n}{\lambda}.$$

The Cramér–Rao bound for an unbiased estimator of λ is $1/I_n(\lambda) = \lambda/n$. Because $\text{Var}(\bar{X}) = \lambda/n$, $\bar{X}$ attains the bound and is efficient.

[7 pts]

(d) Conditional on $T = t$, $(X_1, \dots, X_n)$ is multinomial with t trials and cell probabilities $1/n$. Hence

$$X_1 \mid T = t \sim \text{Binomial}(t, 1/n), \quad \boxed{\mathbb{E}[X_1 \mid T] = T/n}.$$

Rao–Blackwellization turns the rough unbiased estimator X_1 into $\bar{X}$, preserving the mean and weakly reducing variance; here the result is the efficient estimator.

[5 pts]

SOLUTION 3 Conditional Gaussian calculation

Partition

$$\mu_Y = 10, \quad \mu_X = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \Sigma_{YY} = 9, \quad \Sigma_{YX} = \begin{pmatrix} 2 & 1 \end{pmatrix}, \quad \Sigma_{XX} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

The conditional formula is

$$Y \mid \mathbf{X} = \mathbf{x} \sim \mathcal{N}\left(\mu_Y + \Sigma_{YX} \Sigma_{XX}^{-1}(\mathbf{x} - \mu_X), \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}\right).$$

(a) At $\mathbf{x} = (4, 0)^T$, $\mathbf{x} - \mu_X = (2, 1)^T$ and $\Sigma_{YX} \Sigma_{XX}^{-1} = (1, 1)$. Thus

$$\mu_{Y|X} = 10 + (1, 1) \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 13, \quad \sigma_{Y|X}^2 = 9 - (1, 1) \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 6.$$

Therefore

$$Y \mid (X_1 = 4, X_2 = 0) \sim \mathcal{N}(13, 6).$$

[8 pts]

(b) The residual variance fraction is $6/9 = 2/3$, so the explained fraction is $1 - 2/3 = 1/3$.

[4 pts]

(c) A local nonparametric estimate needs observations near the conditioning point. A zero-count five-dimensional cell has no empirical conditional denominator and no local response values; a compact-support kernel likewise has no local mass. A Gaussian model estimates a global mean and covariance from the full sample, then evaluates the resulting regression and Schur complement at any $\mathbf{x}$. It is therefore defined at the empty slice, though its credibility comes from the parametric structure rather than local observations.

[4 pts]

SOLUTION 4 Corrected Second District Bayesian update

(a) The historical two-party share is

$$\hat{p}_{2020} = \frac{154,377}{154,377 + 176,468} = 0.4666143.$$

An effective sample size of 50 means $\alpha_0 + \beta_0 = 50$ and $\alpha_0/(\alpha_0 + \beta_0) = \hat{p}_{2020}$. Therefore

$$\alpha_0 = 50\hat{p}_{2020} = 23.3307, \quad \beta_0 = 50(1 - \hat{p}_{2020}) = 26.6693.$$

[5 pts]

(b) The binomial likelihood kernel from 210 Trump and 270 Harris respondents is $p^{210}(1-p)^{270}$. Multiplying by the Beta prior adds the corresponding pseudo-counts:

$$p \mid \text{data} \sim \text{Beta}(23.3307 + 210, 26.6693 + 270) = \text{Beta}(233.3307, 296.6693).$$

[5 pts]

(c) The posterior effective sample size is $233.3307 + 296.6693 = 530$. The posterior mean is

$$\mathbb{E}[p \mid \text{data}] = \frac{233.3307}{530} = 0.4402466.$$

By posterior prediction for one Bernoulli observation, the probability that the next sampled voter supports Trump is the same posterior mean, 0.4402466.

[5 pts]

(d) Exactly,

$$P(p > 0.5 \mid \text{data}) = 1 - I_{0.5}(233.3307, 296.6693),$$

where $I_x(a, b)$ is the regularized incomplete beta function. Using the supplied posterior standard deviation,

$$z = \frac{0.5 - 0.44025}{0.02154} \approx 2.77,$$

so the normal approximation is $1 - \Phi(2.77) = 0.0028$. The exact Beta tail is 0.002919. **[5 pts]**

(e) The superseded poll had a Trump sample majority, so it moved the prior above 0.5 and produced mean 0.5422. The corrected poll has Trump share $210/480 = 0.4375$, a Harris majority that dominates the modest 50-observation prior and moves the posterior mean to 0.44025. Thus the posterior district-win probability changes from approximately 0.9864 under the typo to approximately 0.002919 under the correction. **[4 pts]**

CORRECTION AUDIT – NOT A TIMED CALCULATION

Using the same independent-district Beta model and fixed 2020 district turnout weights as Homework 3, the corrected counts make four Trump electoral votes the modal Nebraska outcome. The deterministic integration below is supplied as an audit; the mock does not ask you to reproduce it by hand.

Quantity	Distribution	Mean	$P(p > 0.5)$
Prior D2	Beta(23.3307, 26.6693)	0.466614	–
Old typo posterior	Beta(370.3307, 312.6693)	0.542212	0.986413
Corrected posterior	Beta(233.3307, 296.6693)	0.440247	0.002919

Trump EV V	$P(V \mid \text{corrected data})$	Percentage
0	0.000000857	0.0000857%
1	0.000105145	0.0105145%
2	0.000286052	0.0286052%
3	0.024942571	2.4942571%
4	0.971819507	97.1819507%
5	0.002845867	0.2845867%

Under this stated model, the statewide win probability is 0.999608 and the observed four-electoral outcome has probability about 97.18%. This replaces the earlier, erroneous conclusion that four electors were a roughly 3.80% event.

SOLUTION 5 Gamma–Poisson posterior prediction

Let $\mathcal{D} = (x_1, \dots, x_n)$, let $S = \sum_{i=1}^n x_i$, and use the shape–rate convention

$$p(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}, \quad \lambda > 0.$$

Here $a = 2$, $b = 1$, $n = 4$, and $S = 4 + 2 + 3 + 5 = 14$.

(a) Conditional independence gives the joint likelihood

$$\begin{aligned} L(\lambda; \mathcal{D}) &= \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \\ &= \frac{1}{\prod_{i=1}^n x_i!} \lambda^{\sum_i x_i} e^{-n\lambda} \propto \lambda^S e^{-n\lambda}. \end{aligned}$$

Bayes' rule therefore gives

$$\begin{aligned} p(\lambda \mid \mathcal{D}) &\propto L(\lambda; \mathcal{D})p(\lambda) \\ &\propto \lambda^S e^{-n\lambda} \lambda^{a-1} e^{-b\lambda} \\ &= \lambda^{a+S-1} e^{-(b+n)\lambda}. \end{aligned}$$

This is the kernel of a Gamma distribution. Restoring its normalizing constant,

$$p(\lambda \mid \mathcal{D}) = \frac{(b+n)^{a+S}}{\Gamma(a+S)} \lambda^{a+S-1} e^{-(b+n)\lambda},$$

so

$$\lambda \mid \mathcal{D} \sim \text{Gamma}(a+S, b+n) = \text{Gamma}(16, 5).$$

The posterior shape is the prior shape plus the total observed count, while the posterior rate is the prior rate plus the number of exposure periods. **[5 pts]**

(b) To derive the posterior variance rather than quote it, write $\alpha = a+S$ and $\beta = b+n$. For any $k > -\alpha$, the k th posterior raw moment is

$$\begin{aligned} \mathbb{E}[\lambda^k \mid \mathcal{D}] &= \int_0^\infty \lambda^k \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} d\lambda \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty \lambda^{\alpha+k-1} e^{-\beta\lambda} d\lambda. \end{aligned}$$

Set $t = \beta\lambda$, so that $\lambda = t/\beta$ and $d\lambda = dt/\beta$. Then

$$\begin{aligned} \mathbb{E}[\lambda^k \mid \mathcal{D}] &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty \left(\frac{t}{\beta}\right)^{\alpha+k-1} e^{-t} \frac{dt}{\beta} \\ &= \frac{1}{\beta^k \Gamma(\alpha)} \int_0^\infty t^{\alpha+k-1} e^{-t} dt \\ &= \frac{\Gamma(\alpha+k)}{\Gamma(\alpha) \beta^k}. \end{aligned}$$

Taking $k = 1$ and using $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ gives

$$\mathbb{E}[\lambda \mid \mathcal{D}] = \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)\beta} = \frac{\alpha}{\beta}.$$

Taking $k = 2$ and using $\Gamma(\alpha+2) = \alpha(\alpha+1)\Gamma(\alpha)$ gives

$$\mathbb{E}[\lambda^2 \mid \mathcal{D}] = \frac{\alpha(\alpha+1)}{\beta^2}.$$

Consequently,

$$\begin{aligned} \text{Var}(\lambda \mid \mathcal{D}) &= \mathbb{E}[\lambda^2 \mid \mathcal{D}] - (\mathbb{E}[\lambda \mid \mathcal{D}])^2 \\ &= \frac{\alpha(\alpha+1)}{\beta^2} - \frac{\alpha^2}{\beta^2} = \boxed{\frac{\alpha}{\beta^2}}. \end{aligned}$$

For $(\alpha, \beta) = (16, 5)$,

$$\mathbb{E}[\lambda \mid \mathcal{D}] = \frac{16}{5}, \quad \mathbb{E}[\lambda^2 \mid \mathcal{D}] = \frac{272}{25}, \quad \text{Var}(\lambda \mid \mathcal{D}) = \frac{16}{25}.$$

Finally, the posterior log density is $\log p(\lambda \mid \mathcal{D}) = 15 \log \lambda - 5\lambda + C$. Its derivative $15/\lambda - 5$ vanishes at $\lambda = 3$, and its second derivative $-15/\lambda^2$ is negative, so the posterior mode is 3. **[4 pts]**

(c) For $y = 0, 1, 2, \dots$,

$$\begin{aligned} P(Y = y \mid \text{data}) &= \int_0^\infty \frac{e^{-\lambda} \lambda^y}{y!} \frac{5^{16}}{\Gamma(16)} \lambda^{15} e^{-5\lambda} d\lambda \\ &= \frac{\Gamma(y+16)}{y! \Gamma(16)} \left(\frac{5}{6}\right)^{16} \left(\frac{1}{6}\right)^y. \end{aligned}$$

This is a negative-binomial pmf with $r = 16$ and success-probability convention $q = 5/6$, where Y counts failures before r successes. **[7 pts]**

(d) The posterior predictive experiment has two stages. After the observed data have been fixed, first draw a plausible rate $\lambda \sim \text{Gamma}(16, 5)$ from its posterior; then draw the future count $Y \mid \lambda \sim \text{Poisson}(\lambda)$. Conditional on λ , the future observation is independent of the old sample, so

$$\mathbb{E}[Y \mid \lambda, \text{data}] = \lambda, \quad \text{Var}(Y \mid \lambda, \text{data}) = \lambda.$$

Predictive mean. The law of total expectation averages the conditional Poisson mean over all rates still plausible under the posterior:

$$\begin{aligned} \mathbb{E}[Y \mid \text{data}] &= \mathbb{E}_{\lambda \mid \text{data}}[\mathbb{E}[Y \mid \lambda, \text{data}]] \\ &= \mathbb{E}[\lambda \mid \text{data}] = \frac{16}{5} = 3.2. \end{aligned}$$

Thus the Bayes point prediction under squared-error loss is 3.2. This need not be an integer: it is the long-run average of the predictive distribution, not a claim that a single count can equal 3.2.

Predictive variance. The law of total variance separates the two levels of randomness:

$$\begin{aligned} \text{Var}(Y \mid \text{data}) &= \mathbb{E}_{\lambda \mid \text{data}}[\text{Var}(Y \mid \lambda, \text{data})] + \text{Var}_{\lambda \mid \text{data}}(\mathbb{E}[Y \mid \lambda, \text{data}]) \\ &= \mathbb{E}[\lambda \mid \text{data}] + \text{Var}(\lambda \mid \text{data}). \end{aligned}$$

For a shape–rate $\text{Gamma}(\alpha, \beta)$ distribution, $\mathbb{E}[\lambda] = \alpha/\beta$ and $\text{Var}(\lambda) = \alpha/\beta^2$. Hence

$$\text{Var}(Y \mid \text{data}) = \frac{16}{5} + \frac{16}{25} = \frac{96}{25} = 3.84.$$

Source of predictive uncertainty	Mathematical term	Value
Future Poisson variation, present even if λ were known	$\mathbb{E}[\text{Var}(Y \mid \lambda, \text{data}) \mid \text{data}]$	3.20
Posterior uncertainty about the unknown rate λ	$\text{Var}(\mathbb{E}[Y \mid \lambda, \text{data}] \mid \text{data})$	0.64
Total posterior predictive variance	Sum of the two terms	3.84

The additivity is exact. It does not require the two sources to be independent: the residual $Y - \mathbb{E}[Y \mid \lambda] = Y - \lambda$ is orthogonal to every function of λ , because its conditional mean given λ is zero. Equivalently, if $m = \mathbb{E}[Y \mid \text{data}]$, then

$$Y - m = (Y - \lambda) + (\lambda - m),$$

and the cross term has expectation zero after conditioning on λ .

The same identity can be derived without invoking the named variance law. For a Poisson variable,

$$\mathbb{E}[Y^2 \mid \lambda, \mathcal{D}] = \text{Var}(Y \mid \lambda, \mathcal{D}) + (\mathbb{E}[Y \mid \lambda, \mathcal{D}])^2 = \lambda + \lambda^2.$$

Applying total expectation to this second moment gives

$$\mathbb{E}[Y^2 \mid \mathcal{D}] = \mathbb{E}[\lambda \mid \mathcal{D}] + \mathbb{E}[\lambda^2 \mid \mathcal{D}].$$

Therefore

$$\begin{aligned} \text{Var}(Y \mid \mathcal{D}) &= \mathbb{E}[Y^2 \mid \mathcal{D}] - (\mathbb{E}[Y \mid \mathcal{D}])^2 \\ &= \mathbb{E}[\lambda \mid \mathcal{D}] + \mathbb{E}[\lambda^2 \mid \mathcal{D}] - (\mathbb{E}[\lambda \mid \mathcal{D}])^2 \\ &= \mathbb{E}[\lambda \mid \mathcal{D}] + \text{Var}(\lambda \mid \mathcal{D}). \end{aligned}$$

Numerically, this is

$$\frac{16}{5} + \frac{272}{25} - \left(\frac{16}{5}\right)^2 = \frac{80 + 272 - 256}{25} = \frac{96}{25}.$$

Finally, the predictive distribution is overdispersed relative to a single Poisson distribution because

$$\text{Var}(Y \mid \text{data}) - \mathbb{E}[Y \mid \text{data}] = \text{Var}(\lambda \mid \text{data}) = \frac{16}{25} > 0,$$

with dispersion index

$$\frac{\text{Var}(Y \mid \text{data})}{\mathbb{E}[Y \mid \text{data}]} = \frac{96/25}{16/5} = \frac{6}{5} = 1.2.$$

Thus posterior uncertainty increases the Poisson baseline variance by 20%. The same result follows directly from the negative-binomial distribution in part (c): for $r = 16$ and $q = 5/6$, its mean is $r(1 - q)/q = 16/5$ and its variance is $r(1 - q)/q^2 = 96/25$. As the posterior for λ becomes more concentrated, the second variance term shrinks and the predictive law approaches the equidispersion of a Poisson model. **[4 pts]**

Parameterization trap

With a Gamma rate b , the update is $b + n$ and the mean is a/b . If a source uses a scale ϑ , the update must first be translated; adding n directly to a scale is wrong.

GENERAL GAMMA–POISSON PREDICTIVE RULE

Whenever

$$\lambda \mid \text{data} \sim \text{Gamma}(\alpha, \beta) \quad \text{in the shape–rate convention,} \quad Y \mid \lambda \sim \text{Poisson}(\lambda),$$

integrating out λ gives

$$Y \mid \text{data} \sim \text{NegBin}\left(r = \alpha, q = \frac{\beta}{\beta + 1}\right),$$

under the convention that Y counts failures before r successes. Its moments can be written directly in posterior terms:

$$\mathbb{E}[Y \mid \text{data}] = \frac{\alpha}{\beta}, \quad \text{Var}(Y \mid \text{data}) = \frac{\alpha}{\beta} + \frac{\alpha}{\beta^2} = \frac{\alpha(\beta + 1)}{\beta^2}.$$

Because $\alpha = a + S$ and $\beta = b + n$, the entire conjugate chain can also be recorded in the original prior-and-data symbols:

$$\text{Var}(\lambda \mid \mathcal{D}) = \frac{a + S}{(b + n)^2}, \quad \mathbb{E}[Y \mid \mathcal{D}] = \frac{a + S}{b + n}, \quad \text{Var}(Y \mid \mathcal{D}) = \frac{(a + S)(b + n + 1)}{(b + n)^2}.$$

Therefore the predictive dispersion index is $1 + 1/\beta$. For the present posterior, $(\alpha, \beta) = (16, 5)$, so it is $6/5 = 1.2$.

Why a plug-in Poisson prediction is too narrow

Replacing the unknown rate by its posterior mean would give $Y_{\text{plug}} \sim \text{Poisson}(16/5)$. This shortcut preserves the predictive mean but forces the variance to equal $16/5$, discarding the additional $16/25$ contributed by posterior uncertainty. The exact posterior predictive distribution is therefore wider than the plug-in Poisson law.

Exam-ready derivation in five moves

State the two conditional moments of the sampling model; apply total expectation; apply total variance; substitute the posterior mean and variance; then identify the excess term $\text{Var}(\lambda \mid \text{data})$ and interpret it. This sequence earns method credit even if the last arithmetic step is missed.

Score Diagnosis and Three-Day Plan

Interpret the score only after checking whether each derivation can be reproduced from a blank page.

Score	Interpretation	Next move
90–100	Exam-ready chain control	Redo every non-full-credit subpart once; spend the remaining time on speed and notation.
75–89	One or two weak threads	Repair the first broken step, then re-solve the whole problem without viewing the key.
60–74	Several derivations are not yet automatic	Rebuild Problems 2, 4, and 5 first; they carry the densest inference logic.
Below 60	Recall is still fragmented	Copy no solutions. Read one solution, close it, reconstruct it, and compare line by line.

Error log

Failure type	What it looks like	Repair
Model/support	Forgot support or used the wrong likelihood	Rewrite the model before algebra.
Kernel/family	Could not recognize a sufficient statistic or conjugate kernel	Circle all parameter-dependent terms.
Algebra	Right method, arithmetic or derivative failed	Redo slowly, then once at speed.
Interpretation	Correct formula, wrong probabilistic meaning	Say the answer aloud in one sentence.
Correction control	Used 347/286 anywhere	Replace with 210/270 and restart the update.

Monday, 28 September: Diagnose and repair

- Sit the complete 90-minute mock under the announced exam conditions.
- Grade for method. Mark the first invalid line, not merely the final wrong number.
- Re-solve only the missed subparts after a 20-minute break. End by writing from memory the six update formulas on the final page of this packet.

Tuesday, 29 September: Classical inference plus the correction

- 45 minutes: Pareto likelihood, support, MLE, and the likelihood–KL identity.
- 45 minutes: Poisson factorization, Rao–Blackwell, Fisher information, and CRLB.
- 35 minutes: derive the corrected Second District posterior from the raw counts 210 and 270. Do not begin from the answer.

Wednesday, 30 September: Bayesian consolidation and taper

- 45 minutes: conditional Gaussian formula, Beta–Binomial, Gamma–Poisson, and posterior prediction from a blank page.
- 35 minutes: re-solve your two lowest-confidence mock problems under time pressure.
- 20 minutes: oral check. Explain why each posterior parameter changes, why predictive variance decomposes, and why the corrected poll reverses the conclusion.
- Stop heavy work early enough to protect sleep; do not add a new topic on Wednesday night.

Thursday, 1 October: Execution

- Arrive at least ten minutes early, as requested by the instructor.
- Bring no notes or electronic device into use. On receiving the exam, scan all questions, mark conjugate-family and support traps, and allocate time by points.
- If stuck, write the parameter-dependent kernel. It usually reveals the statistic, estimator, or posterior family.

Closed-Book Recall Map

WRITE THESE FROM MEMORY – DO NOT BRING THIS PAGE INTO THE EXAM

The exam announcement permits no materials or devices. This page is a rehearsal target for 28–30 September, not an exam aid.

Thread	Memory skeleton
MLE	$L(\theta)$; support; $\ell(\theta)$; score = 0; curvature or boundary; interpret.
Factorization	$f(\mathbf{x} \mid \theta) = g_\theta(T(\mathbf{x}))h(\mathbf{x}) \Rightarrow T$ sufficient.
Fisher / CRLB	$I_n(\theta) = nI_1(\theta)$; $\text{Var}(U) \geq [g'(\theta)]^2 / I_n(\theta)$.
Rao–Blackwell	$U^* = \mathbb{E}[U \mid T]$; same expectation, variance no larger.
Conditional normal	$\mu_{Y x} = \mu_Y + \Sigma_{YX}\Sigma_{XX}^{-1}(x - \mu_X)$; $\Sigma_{Y X} = \Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY}$.
Beta–Binomial	Beta(α, β) + s successes, f failures $\Rightarrow$ Beta($\alpha + s, \beta + f$).
Gamma–Poisson	Gamma(a, b rate) + n counts, total $S \Rightarrow$ Gamma($a + S, b + n$).
Poisson prediction	Gamma posterior mixed with Poisson $\Rightarrow$ negative binomial; $\text{Var}(Y \mid x) = \mathbb{E}[\lambda \mid x] + \text{Var}(\lambda \mid x)$.
Likelihood / KL	$\ell(\theta)/n = -H(\hat{p}) - D(\hat{p} \parallel p_\theta)$.

Last trap scan

Trap	Correct response
Poll correction	Second District: Trump 210, Harris 270; posterior Beta(233.3307, 296.6693).
Beta update	Trump count adds to α ; Harris count adds to β .
Gamma convention	Rate b gives mean a/b ; scale ϑ gives mean $a\vartheta$.
Predictive law	Integrate over the posterior; do not plug in only the posterior mean.
Empirical zero	Unobserved in a small sample is not structurally impossible.
Conditional Gaussian	The variance is a Schur complement and cannot depend on the observed x .

Source note: exam conditions and coverage follow the instructor’s 23 September 2026 announcement; the controlling Homework 3 Problem 2 counts follow the 25 September 2026 correction.